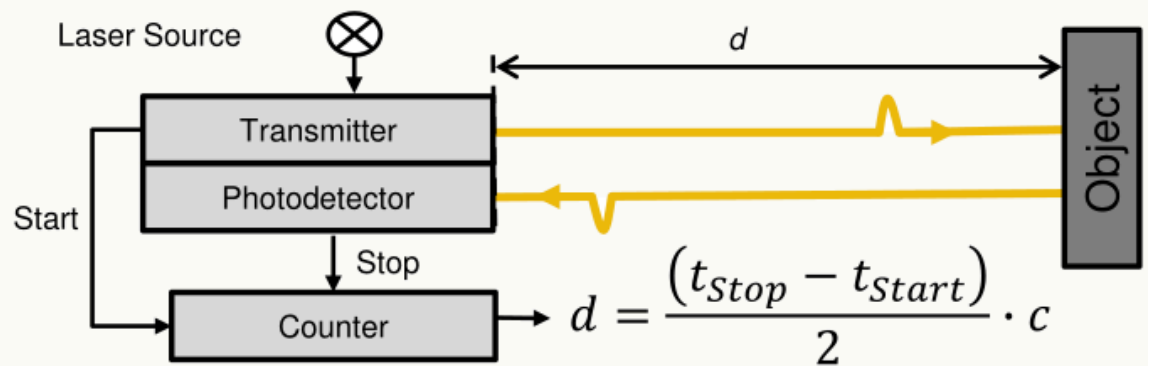
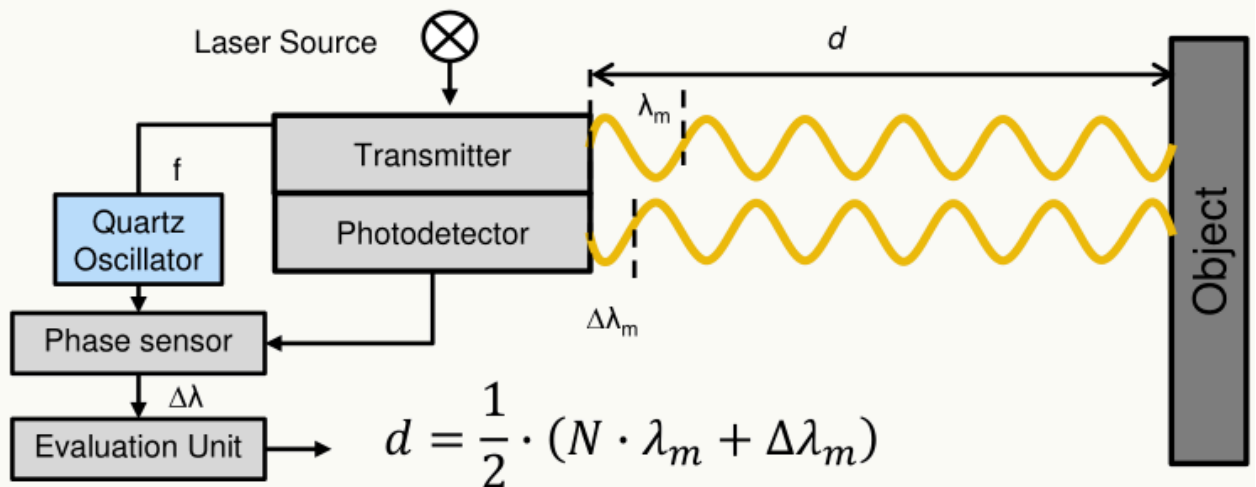


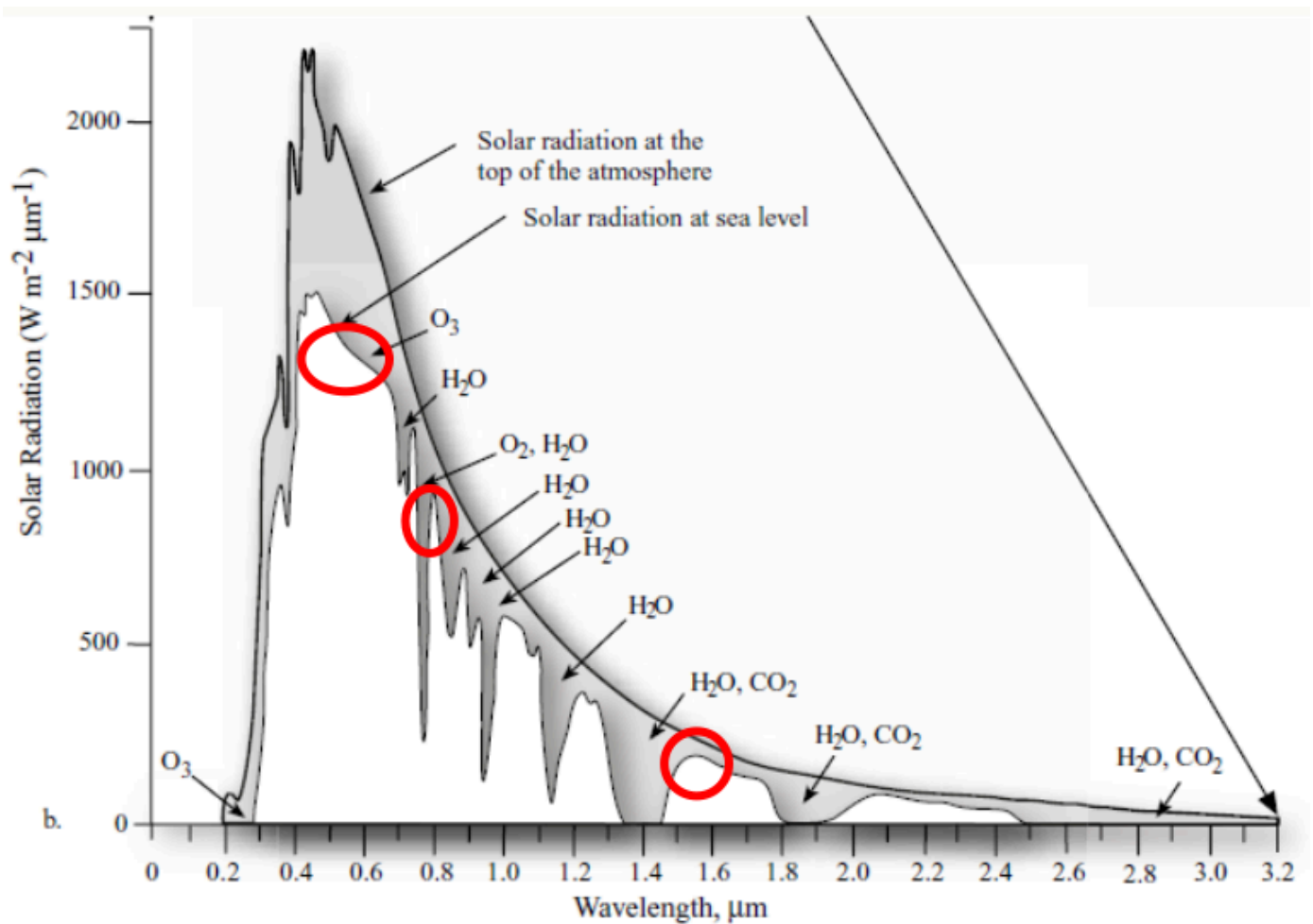
L2: Terrestrial Laser Scanner (TLS) & Errors

Impulse Based

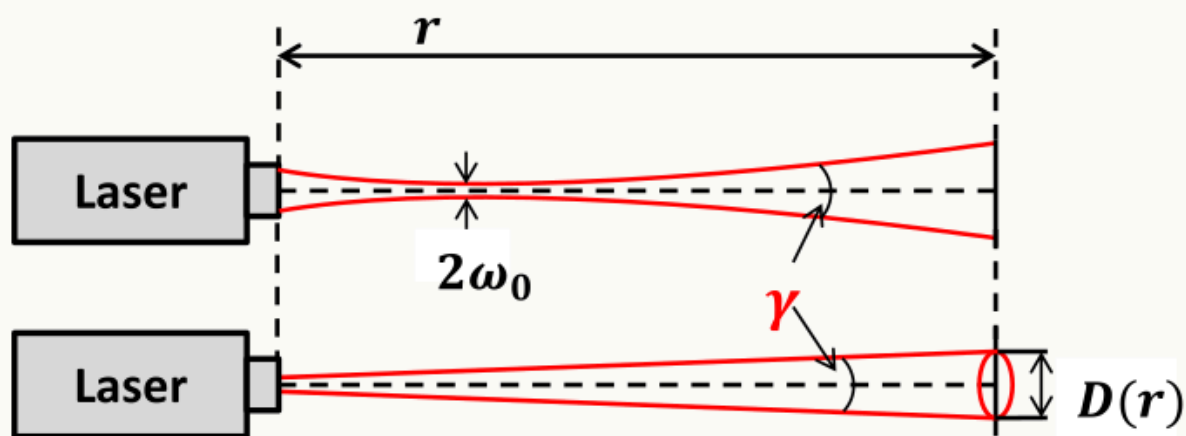


Phase Based (AMCW)





Beam Propagation



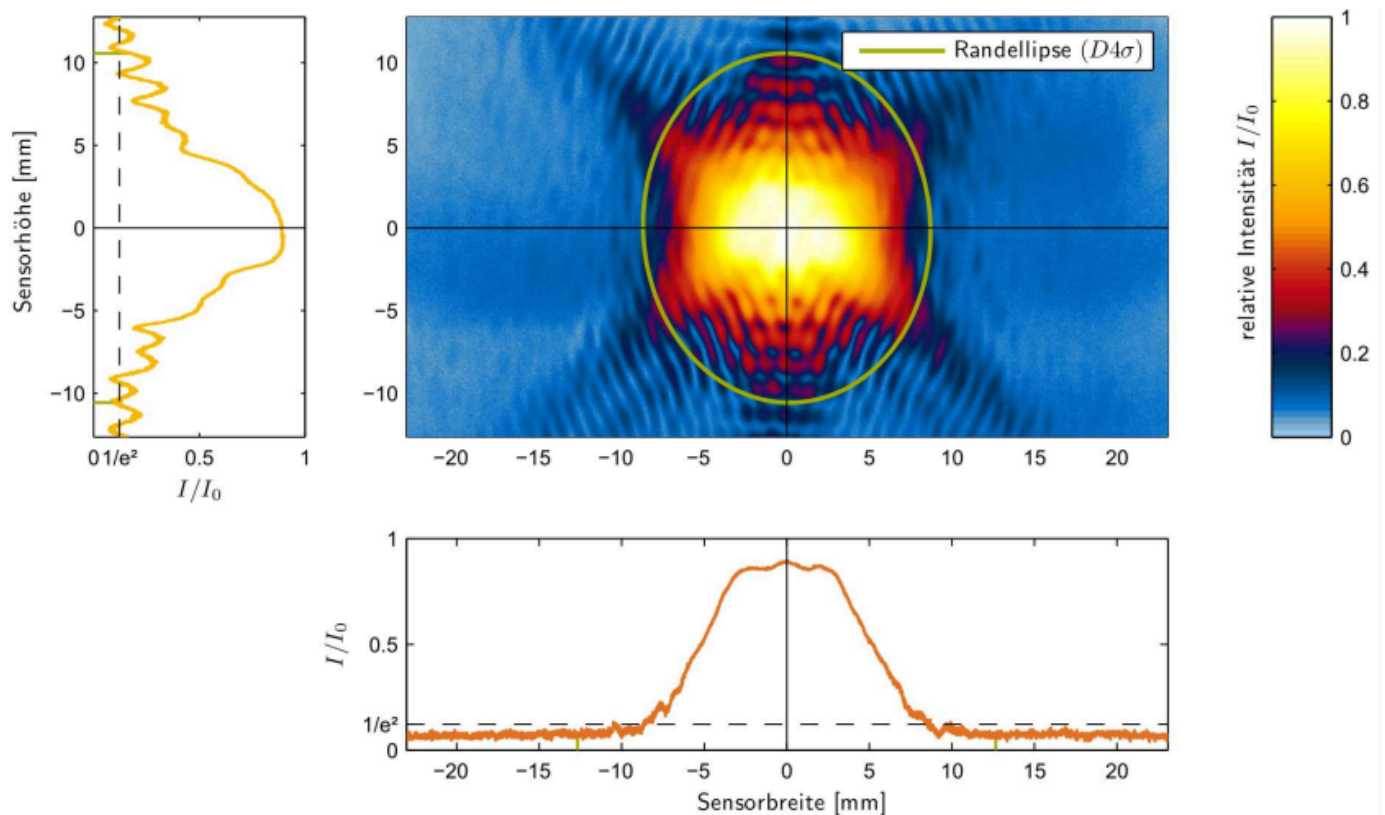
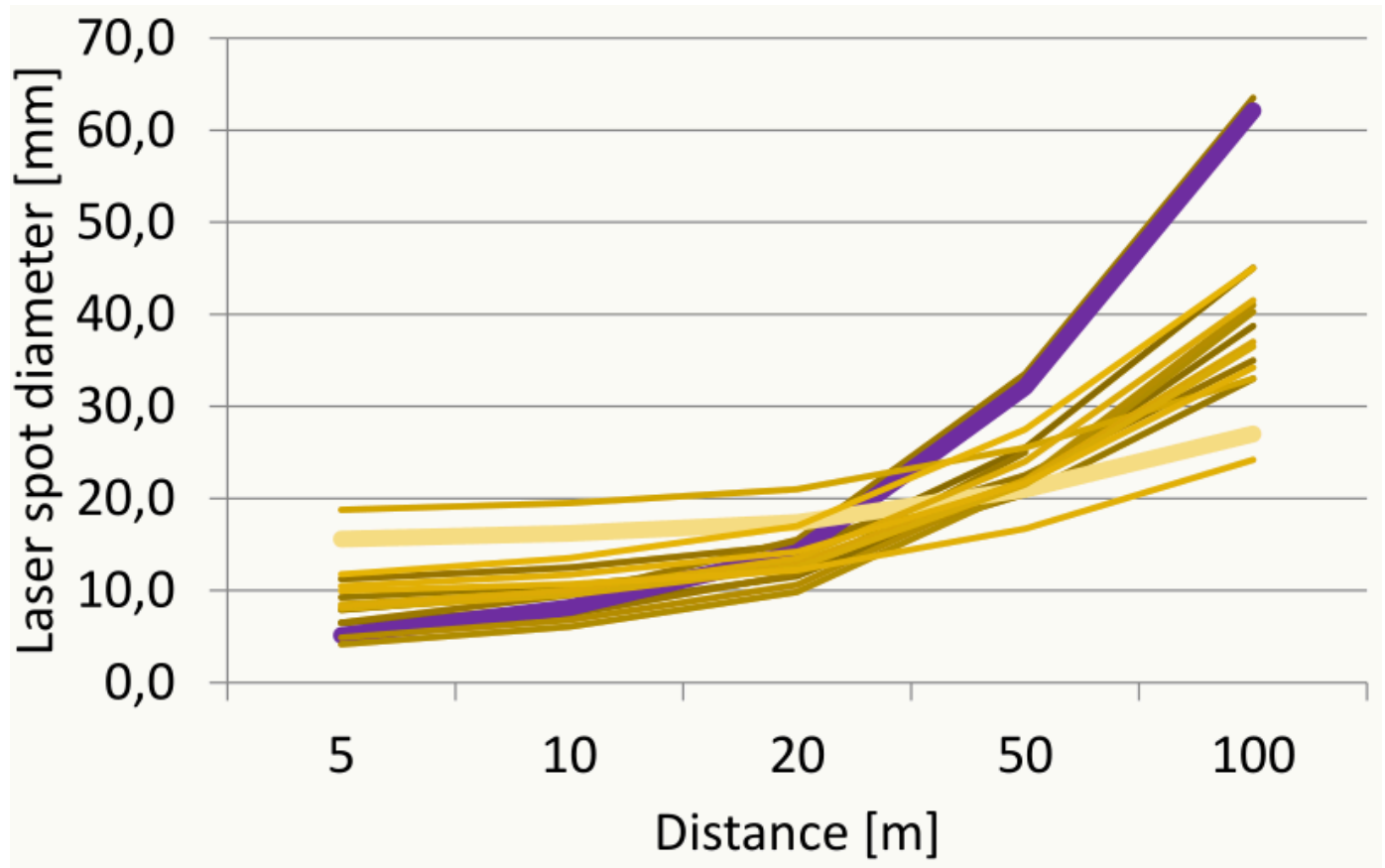
$$D(r) = 2 \cdot \omega_0 \cdot \sqrt{1 + \left(\frac{r \cdot \lambda}{\pi \cdot \omega_0^2} \right)^2}$$

$$\approx 2 \cdot r \cdot \tan\left(\frac{\gamma}{2}\right) + 2 \cdot \omega_0$$

$$\approx r \cdot \gamma + 2 \cdot \omega_0$$

- ω_0 : Radius of beam waist
- λ : Wavelength of laser
- r : distance
- $D(r)$: Footprint
- γ : angle of divergence

Footprints



- Distance measurement effective area:
 - All area patches reflects to scanner
- $\hat{D} = \frac{1}{\int \int_F E(x) dx} \cdot \int \int_F E(X) D(X) dx$

-

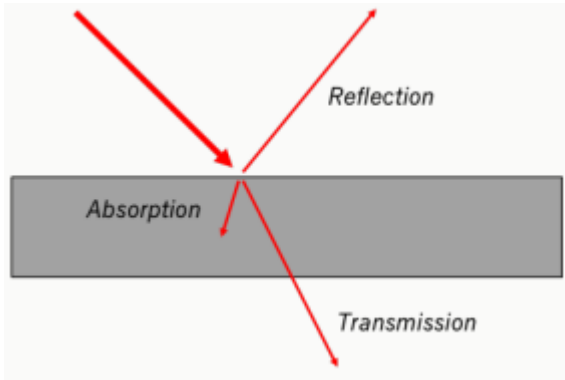
●



●

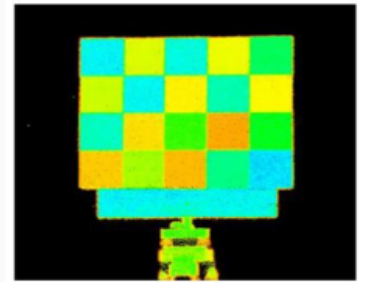
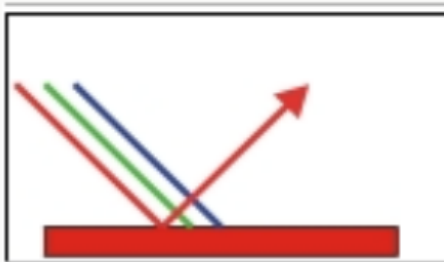


- Influence on received signal strength
- Accuracy (SNR), segmentation/object detection



Reflectance

- Intensity:
 - is measured by the scanner
 - depends on colour and material
 - can be used to segment objects
 - indicates range precision



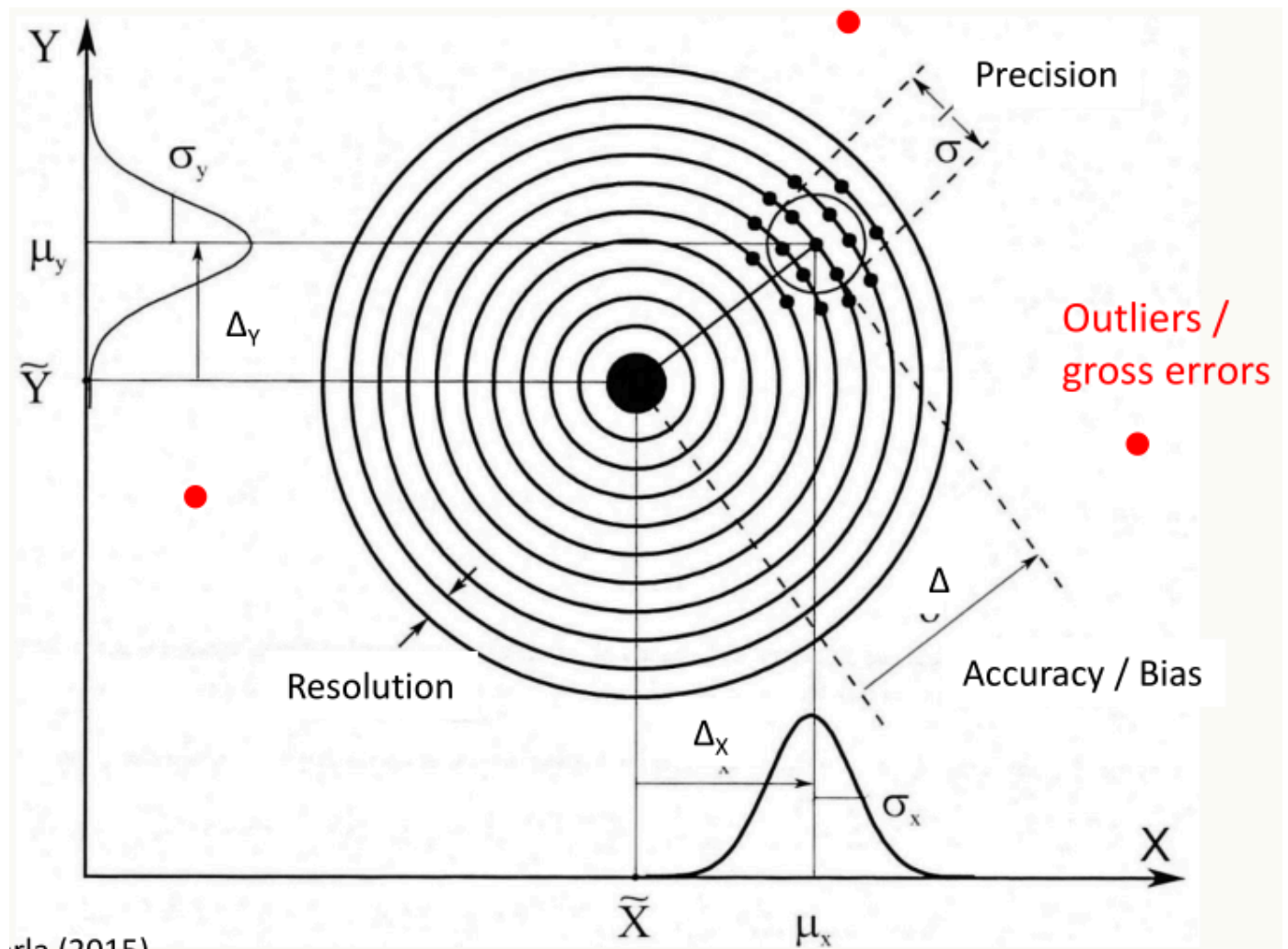
$$P_r = \frac{P_t D_r^2 \eta_{sys}}{4} \eta_{atm} \frac{\cos \alpha}{r^2} \rho$$

Labels and arrows pointing to the equation:

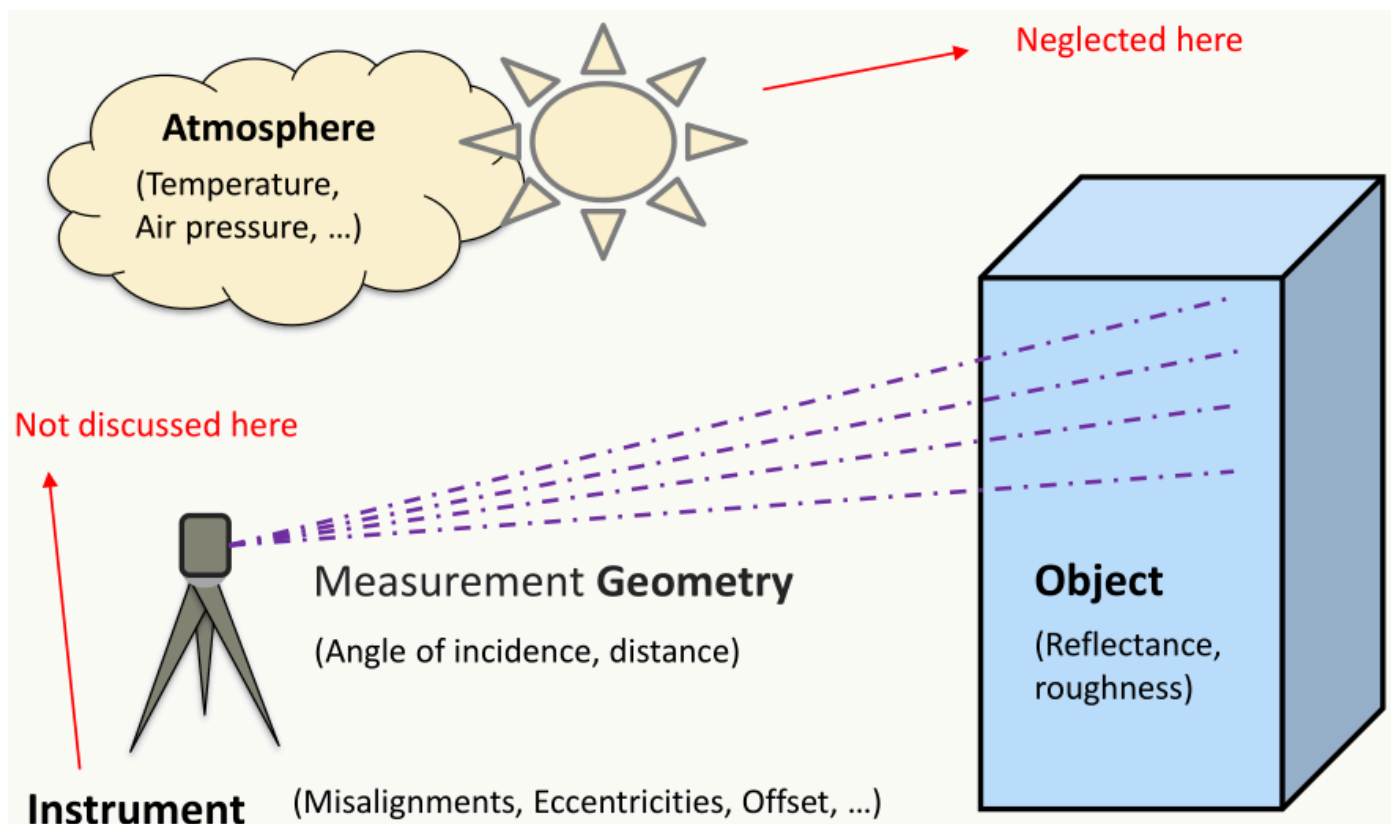
- Transmitted Power** points to P_t
- Receiver Aperture Diameter** points to D_r
- System Transmission Factor** points to η_{sys}
- Received Power** points to P_r
- Atmospheric Transmission Factor** points to η_{atm}
- Angle of Incidence** points to α
- Reflectance of Surface** points to ρ
- Measured Range** points to r

Errors

Types of Errors



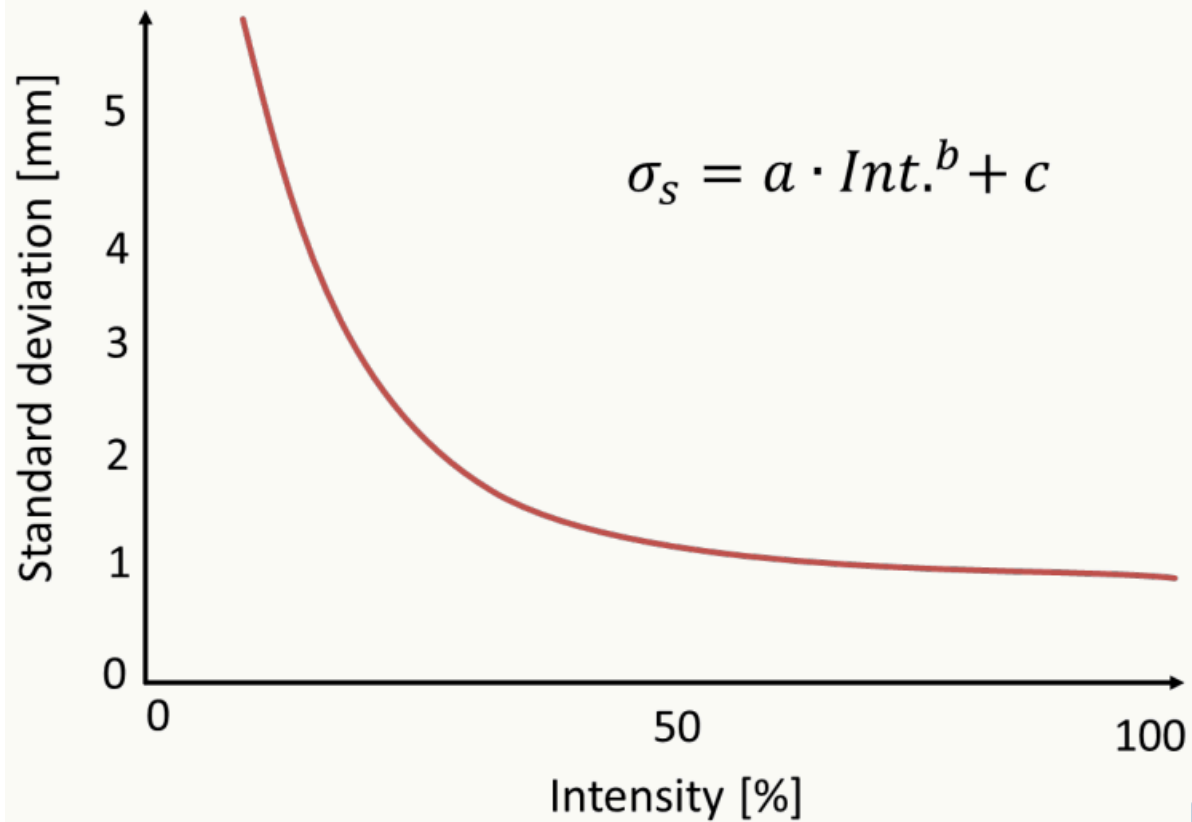
Influences



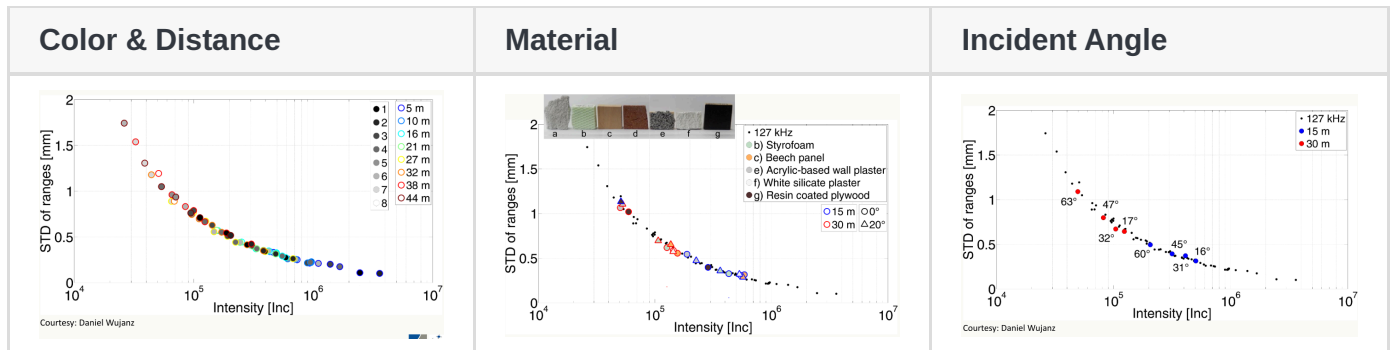
Sources of Error

Category	Element	Random errors	Systematic errors	Outliers/Gross Error
Geometry	Distance	X		
	Incidence angle	x	x	
Object	Color	x	x	
	Material	x	x	Total reflection
	Form		x	Mixed Pixels

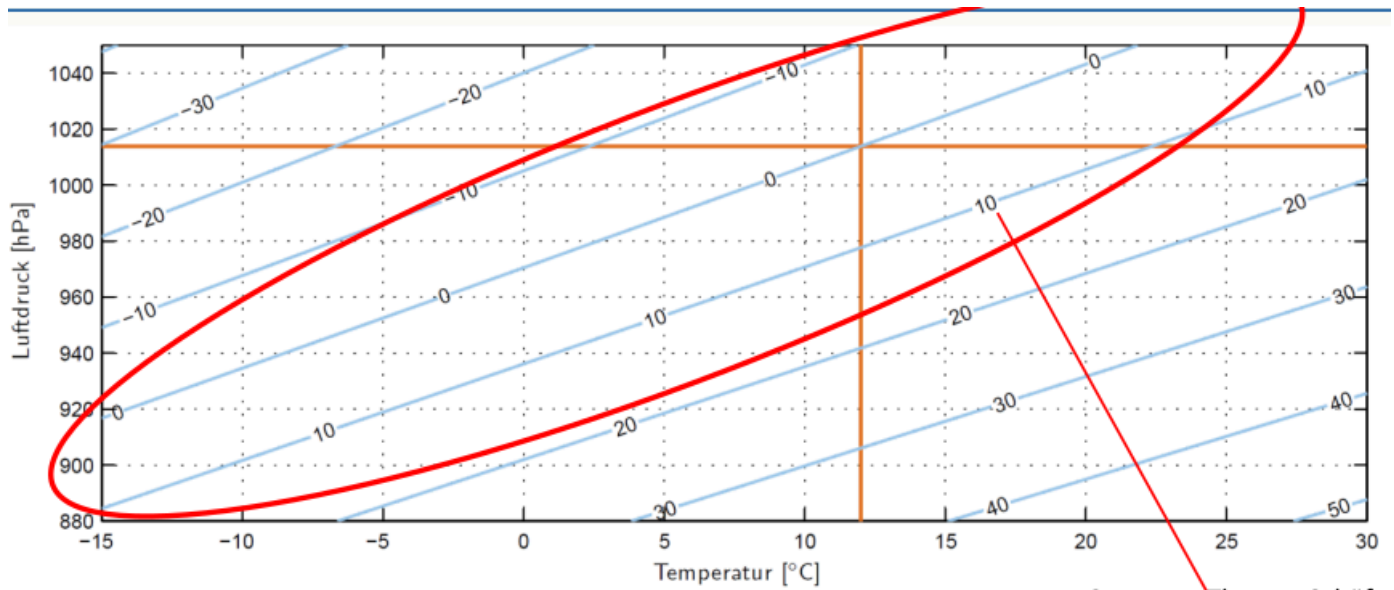
- Intensity as Measure for Precision



Random Errors



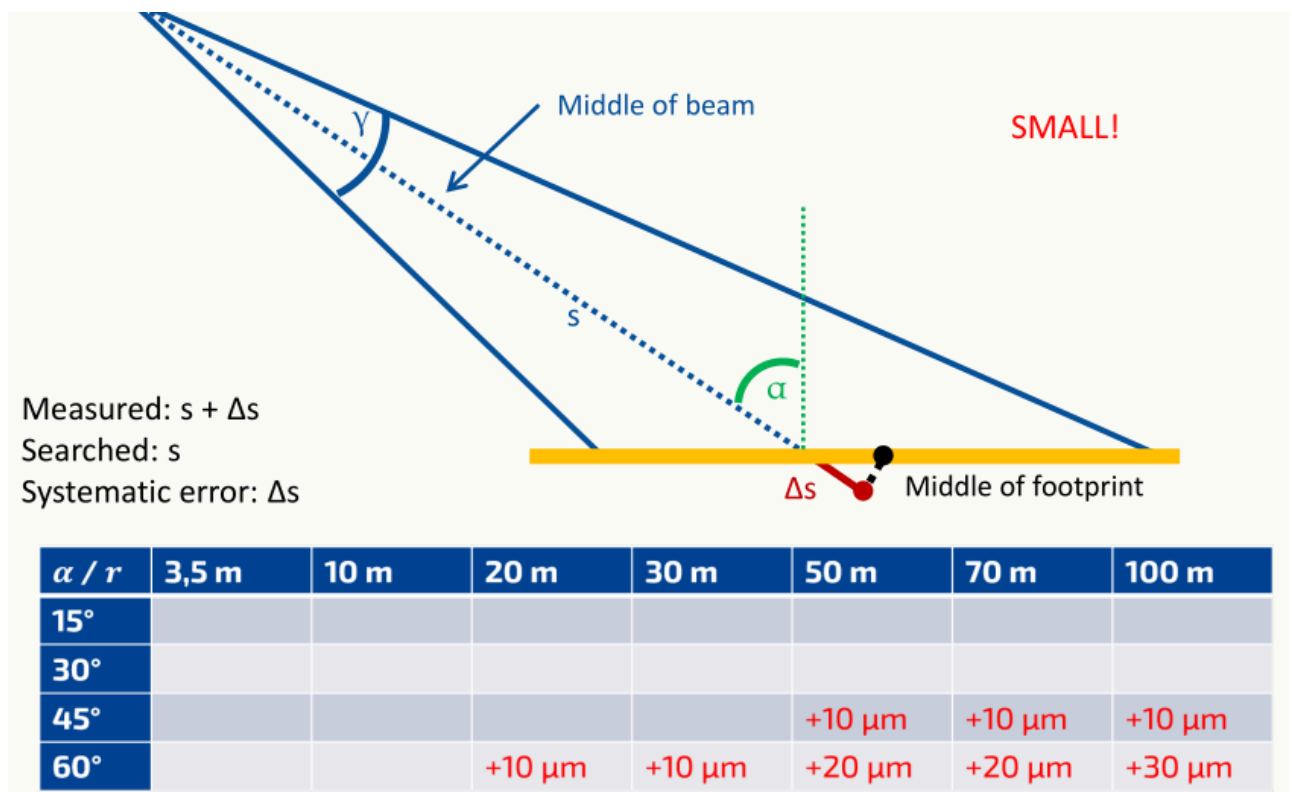
Systematic Errors: Influence of Atmosphere



- $\pm 1\text{mm}@ 100\text{m}$
- Speed of light depends on pressure/temperature
- Atmospheric correction ppm
- Reference Atmosphere: 12°C , 1013.25hPa , 60% Humidity
- Laser wave length 700nm

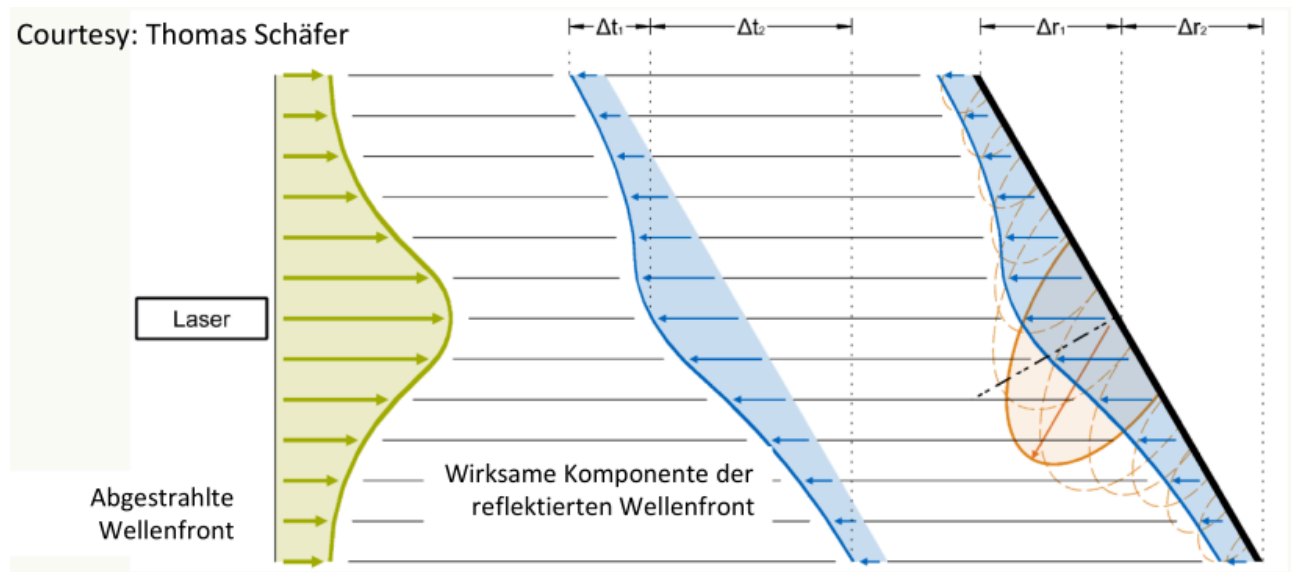
Systematic Errors: Incidence Angle

- Geometric effect

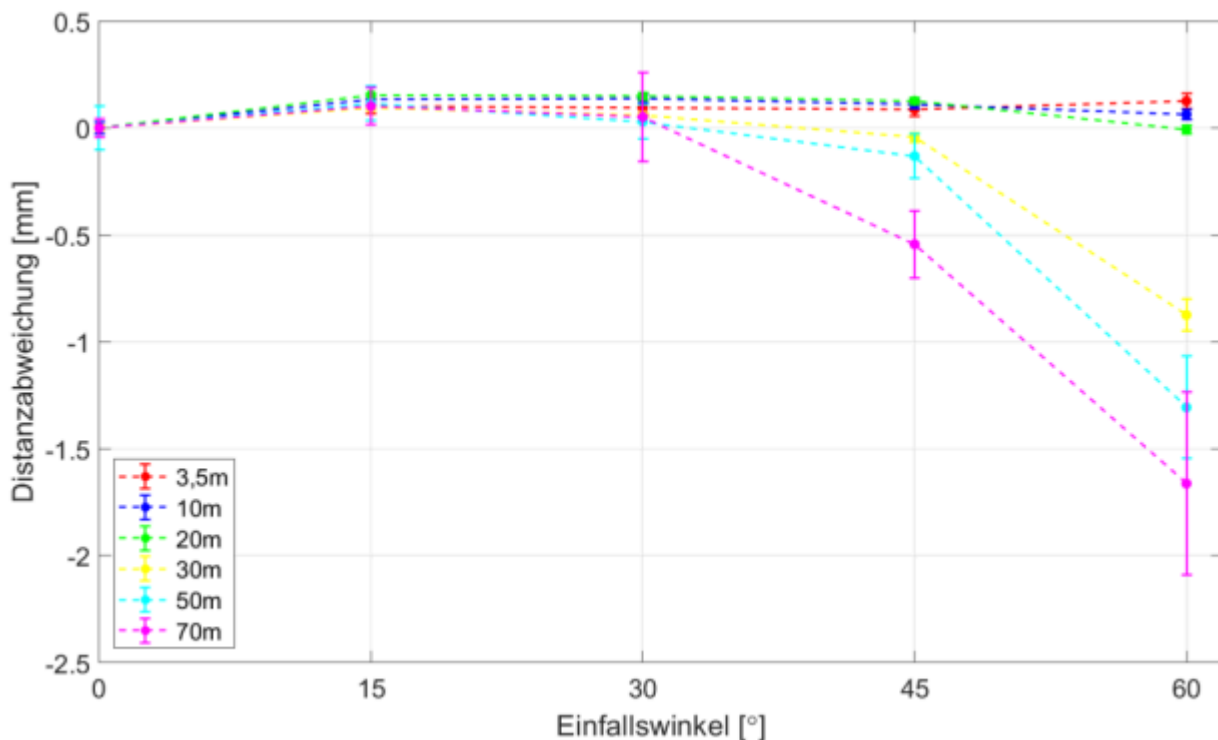


- Physical effect

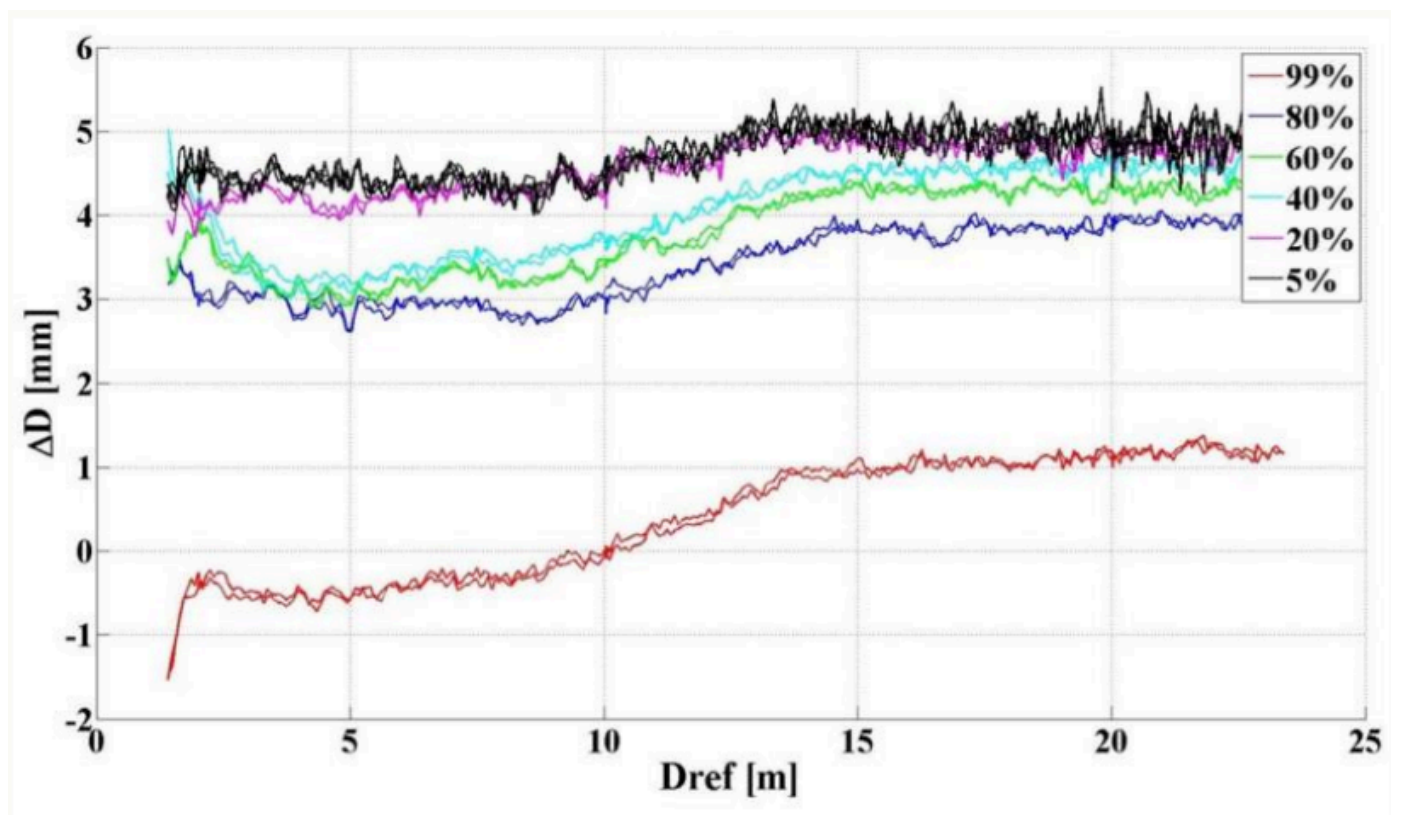
Courtesy: Thomas Schäfer



- Deformation of Wavefront for not perpendicular reflection
- Δr_1 : effective intensities become closer
- Δr_2 : Move away from each other
- Effective intensity distribution at the detector shifted
 - Shorter distance
- Offset: $-1.6mm$ (Z+F Imager 5016)
- Depends on distance $\rightarrow$ Laser spot size
- Depends on incidence angle $\rightarrow$ Laser spot deformation
- **Physical effect is dominant**

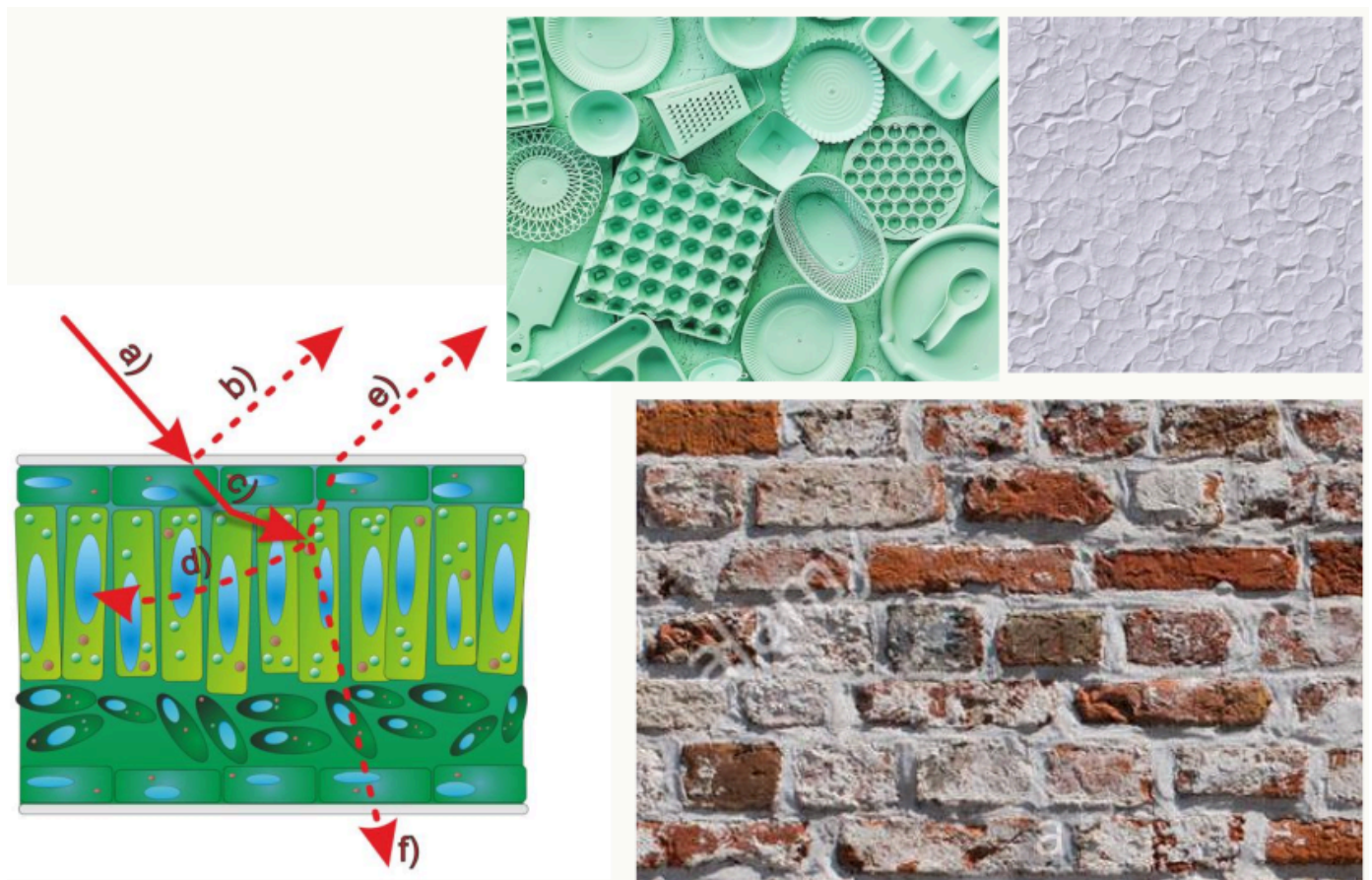


Color/Reflectivity of Target

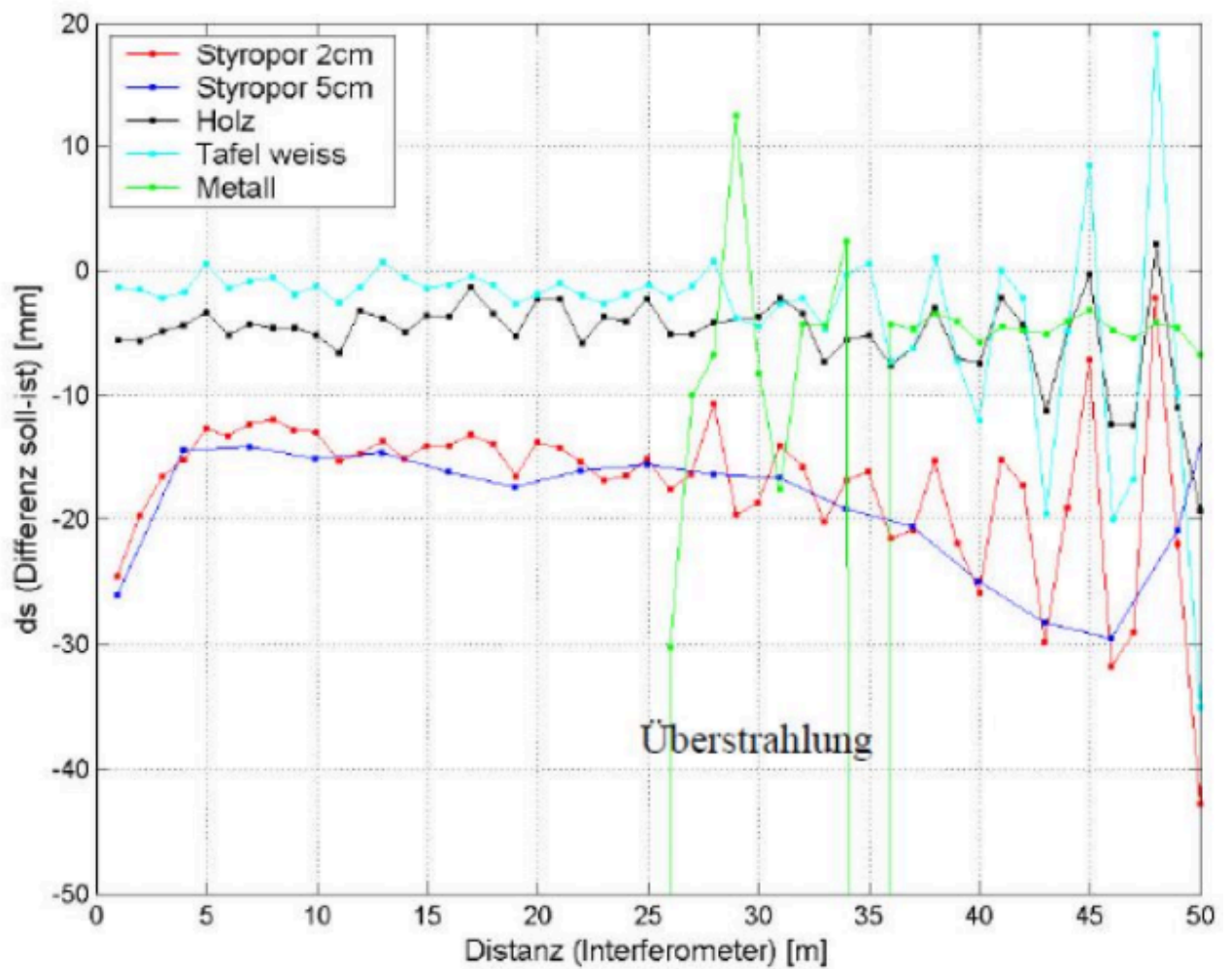


- max 12mm

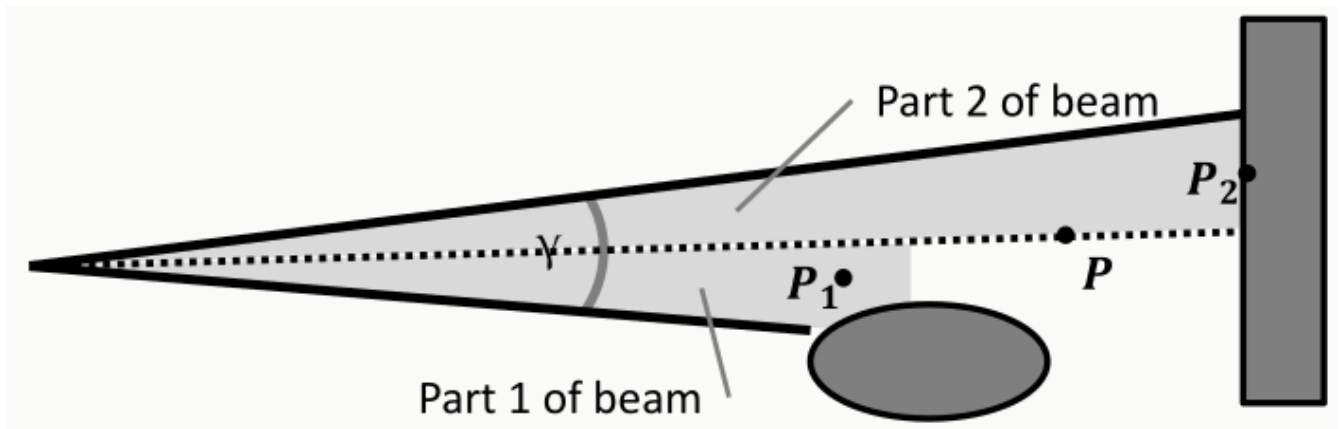
Systematic Error: Material



SYSTEMATIC ERRORS: MATERIAL



Systematic Error: Form



Gross Errors: Total Reflection

